# Do specialized components learn their intended roles— and does this improve forecast uncertainty?

**DecoVAE-FM project** Shared problem: forecast a distribution of future values; test structural meaning, accuracy and calibration together.

## 01 Which assumptions are useful?

Two families address the same task: structured generative models and pretrained forecasting models.

| Work | Assumption and output | Reconstructed evaluation |
|---|---|---|
| **DecoVAE [1]** | Trend + Fourier seasonality; separate VAEs and trend smoothness. Latent forecast samples. | Supervised: 7 datasets; T=96, H=24/720; CRPS / NMAE; 3 retrains. |
| **K²VAE [2]** | Local + global Koopman dynamics; Kalman correction; Gaussian VAE observation model. | ProbTS: 8 short / 9 long; typical T=H=24 or T=96, H≤720. CRPS / NMAE; 5 seeds. |
| **Chronos-2 [3]** | Patches + time / group attention; multivariate and covariate context; quantile forecasts. | Pretrained, zero-shot. fev-bench / GIFT / Chronos-II; rolling origins; WQL / MASE. |
| **Moirai 2.0 [4]** | Channel-independent decoder; multiple future tokens; autoregressive quantile forecasts. | Pretrained, zero-shot. GIFT: 97 tasks, 55 datasets; 9 quantiles; normalized aggregate scores. |
| **FlowState [5]** | S5 state-space encoder; functional Legendre decoder; sampling-rate equivariance; quantiles. | Pretrained, zero-shot. GIFT: 97 tasks; MASE / quantile CRPS; sampling-rate stress tests. |

Why these works? Compare explicit structure, dynamics priors and pretraining, and their alternative uncertainty representations.

## 02 Evidence and its limits

### Structure ≠ recovered roles

A better forecast does not show that decoded trend and seasonality match their intended components. Test roles where the generating process is known.

### CRPS ≠ calibration alone

DecoVAE claims calibration but reports CRPS / NMAE. Add coverage, interval width and PCE (calibration error) by horizon. Adler [6] supplies this lens; it evaluates Chronos-Bolt.

### A leaderboard ≠ a controlled contrast

Different pretraining corpora, contexts, horizons, splits, scales and aggregation can change rankings. ProbTS [7] provides a useful cross-model protocol.

### A shared evaluation contract

Same datasets, context / horizon, rolling origins and normalization. Record pretraining exposure and compute. Report per-dataset scores before pooled ranks. Select variants on validation only.

## 03 Our question → decisive tests

### Current implementation: DecoVAE-FM

Moving-average trend, top-K Fourier seasonality, then a residual. Three Gaussian latent branches; trend + seasonality produce the mean.

**Mean = decoded trend + decoded seasonality**
**Variance head = f(ztrend, zseason, zresidual)**

The residual branch feeds variance; it is not an additive noise path. NLL + KL do not enforce correct roles. “FM” is a transfer ambition, not an established result.

### Retrained 2 × 2 ablation

| Decomposition | Residual on | Residual off |
|---|---|---|
| Current | A | B |
| + trend smoothness | C | D |

A vs B / C vs D: residual contribution.
A vs C / B vs D: structure; compare interactions.
Keep NLL, variance head, sampling and budget.

Add a no-decomposition probabilistic control. Synthetic data: role recovery and variance fit. Real data: accuracy, CRPS, coverage and width.

## 04 Feasible checkpoint and plan

Checkpoint (planned): documented setup + data preparation; audited temporal splits and causal preprocessing; Seasonal Naive; end-to-end smoke test; seeds, metrics and runtime logged; no test tuning.

### Limited local evidence

Saved synthetic diagnostics: periodic signal mainly follows decoded trend; seasonal output is small in 4 examples (posterior-mean decode). This motivates role tests; it does not establish calibration.
This week: baseline + experiment map. Weeks 2–3: 2 × 2, one dataset, 3 seeds. Week 4: second dataset / pretrained baselines as compute allows.

Sources: [1] DecoVAE, arXiv:2608.20052 [2] K²VAE, 2505.23017 [3] Chronos-2, 2510.15821 [4] Moirai 2.0, 2511.11698
[5] FlowState, 2508.05287 [6] Adler et al., 2510.16060 [7] ProbTS, 2310.07446